

Advanced AI for IT

Day 4 | Agents

OQ | Muscat | Wednesday 30 September 2026

Facilitator: Ritesh Vajariya

DAY 4 OF 5

Agents

By 16:00 you will have **measured an agent against a workflow, put five controls on it, and watched a stranger's ticket try to take it over.**

Day 1 Architect

Day 2 Own the model

Day 3 Ground it

Day 4 Agents

Day 5 Integrate and ship

DAY 4 | WORDS WE USE TODAY

Six words, in plain English

Agent

A loop where the model chooses its own next step.

Workflow

Steps you wrote in code. Some of them call a model.

Tool

Something the model can call: look up a ticket, change a ticket.

MCP

A standard plug, so any model can use any tool.

Gate

A rule that checks a change before it happens.

Harness

Everything around the model that makes it safe to run.

DAY 4 | AGENDA

Today in eight blocks

Time	What	How long
08:00	Recap: where your spec stands	15 min
08:15	Tune or retrieve? Settle it on one table	Lab, 40 min
08:55	Agents, then MCP, then two servers on your machine	Talks and demo, to 10:25
10:25	Build four kinds of agent and price them	Lab, 70 min, then a case study
11:50	Dhuhr and lunch	55 min
12:45	Put controls on it	Lab, 90 min
14:30	Attack it	Demo, 30 min
15:00	Integration map, a case study, spec v0.5	To 16:00, break and Asr at 15:15

DAY 4 | IF YOU MISSED SOMETHING

What you need from Days 1 to 3, and what to do if you missed it

You need	If you missed it
Your spec v0.4	Write OPEN in the gaps. Tonight's v0.5 fills them.
The idea that retrieval adds facts and tuning changes behavior	S19 shows it again on one table, in ten minutes.
A working Colab and the shared key	Check now. Ask before 08:15.
Anything you built on Days 2 and 3	Not needed. Labs start from the repo, and most carry a saved run.

DAY 4 | RECAP | 15 MIN

Two minutes each: where your spec stands

Your spec v0.4

Sections 3, 5, 6 and 8, one sentence each.

Your first guess

Tune, retrieve, or both.

Your exposure

The Day 3 failure your system is most exposed to.

DAY 4 | BRIDGE

Until now the model answered. Today it acts.

Days 1 to 3

The model answers. A person decides what happens next.



Day 4

The model looks things up, and can change them. So we add controls.

Tools that write need controls. That is the day.

Run tuning and retrieval against each other on one task set, and read the table

NOTEBOOK	11_three_way: Lab 11, Tune versus retrieve, settled
TIME BOX	25 minutes in the notebook, in a 40-minute slot
RUNS ON	Colab, CPU only. No T4, no download, no credits used. One OpenAI key in Colab Secrets, or it replays the saved run
WORK IN	Your capstone group
IN AZURE	Foundry evaluations score this same table on your own endpoint; Azure AI Search holds the index



Scan to open in Colab

Key: sent to you directly, pasted into Colab Secrets as OPENAI_API_KEY. It is never on screen. Revoked Thursday 16:00.

Before you run: write down your guess

Best routing?

Which of the four arms sends the ticket to the right queue most often?

Best knowledge?

Which arm gets the right fact from the plant's documents?

Does tuning still win on shape?

Day 2's model won on valid JSON. Does it win today?

Two switches, not two answers: tuning changes how the model behaves, retrieval changes what it knows.

Retrieval fixed knowledge. Tuning fixed formatting.

0.42 → 0.86

Knowledge right, with retrieval.

12 → 0

Replies that needed clean-up,
with tuning.

2x the prompt

451 to 918 tokens per ticket. The
price of retrieval.

Schema was 1.00 in every arm. The tuned arm here is a stand-in, so read it as a shape, not a trained adapter.

DAY 4 | S19 TUNE VERSUS RETRIEVE

Retraining is a release. Re-indexing is a file copy.

Change the model

New training set, a run, an eval, a regression check.
Then a release.

Change the documents

Copy the file. Rebuild the index.
The revision filter keeps the old value out.

HSE revises procedures on its own schedule. It does not ask whether your model was retrained.

What kind of failure is it?

Wrong shape

Behavior. Constrain the decoder first. Tune if it persists.

Right shape, wrong number

Knowledge. Retrieval.

Right until the procedure changed

Staleness. Re-index. Retraining will not reach it.

Say it once each: our failure is ____, so we fix it with ____.

DAY 4 | S20 THE AGENT STACK | START HERE

A digital twin question: is compressor KTF-C2 overdue for service?

A model on its own

Answers from training.

It has never seen this plant, this compressor or this week.

What it would need

The last service date.

The vibration trend.

The maintenance procedure.

To answer well, the model has to look something up. That is the whole idea of today.

The same question, three ways

One prompt

The model guesses from training. Fast, cheap, and nothing to check it against.

Workflow

Your code fetches the service record, then asks the model. Steps fixed in advance, by you.

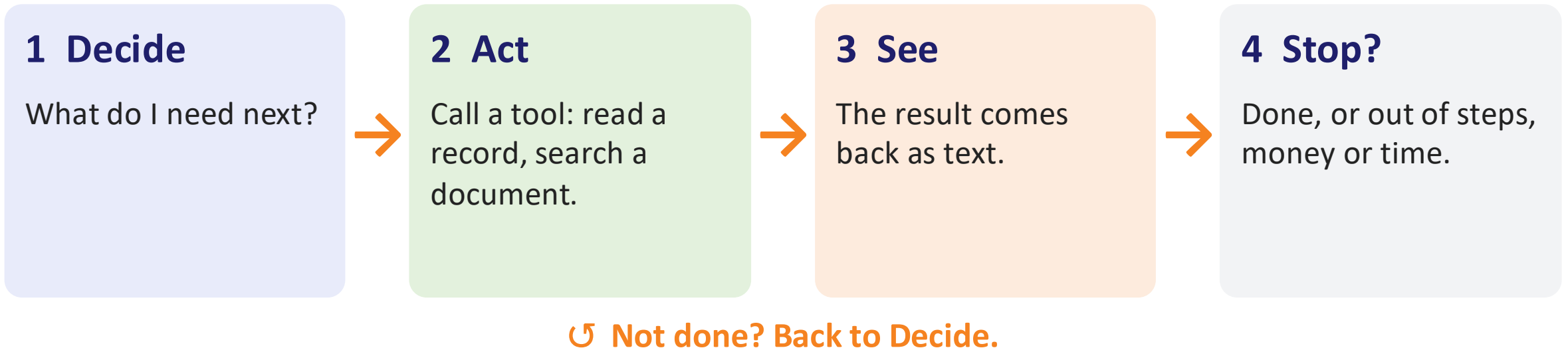
Agent

The model decides to look up the record, sees the trend is missing, looks again, then answers.

Same question. The difference is who chooses the next step.

DAY 4 | S20 THE AGENT STACK

An agent is one loop: decide, act, see the result, repeat



Every agent is this loop, plus tools, plus a rule for when to stop.

DAY 4 | S20 THE AGENT STACK

Three parts: the model decides, the tools do, the harness limits

The model

Decides.

Picks the next step and writes the tool call.

The tools

Do.

Read a record. Search documents. Raise a work order.

The harness

Limits.

Runs the loop, checks each call, counts steps, money and time, writes the log.

You do not build the model. You build the tools and the harness. That is the labs.

DAY 4 | S20 THE AGENT STACK

Why an agent is a different kind of thing to run

Workflow: the path is fixed

You can draw it and test every path.
A wrong answer traces back to a step.

Agent: the path is chosen at runtime

Every turn, the model can pick wrong and act.
More turns: more cost, more time, harder to replay.

The question is never 'agent or not'. It is how far up the ladder you need to go.

DAY 4 | S20 THE AGENT STACK

Three things push you off a single call, and only these three

1 A live fact

Retrieval reads your documents.
It does not know whether ticket
4471 is still open.

2 More than one step

A later step depends on what an
earlier step found.

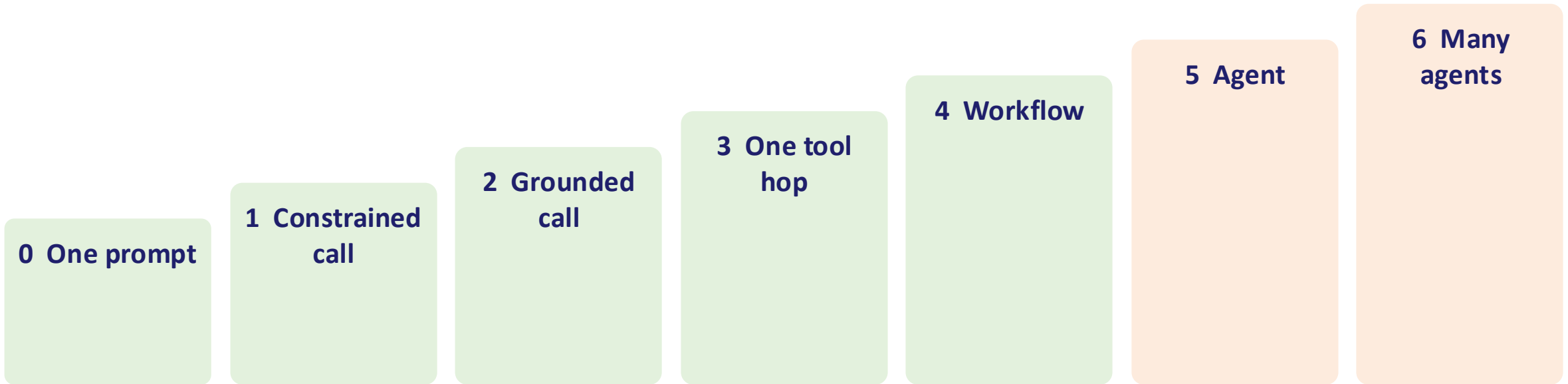
3 Something changes

A ticket updated, a work order
raised, a mail sent.

If none of the three is true, the rest of today is architecture you do not need.

DAY 4 | S20 THE AGENT STACK

Seven rungs. One question separates rung 4 from rung 5.



You decide the next step, in code

The model decides

The line: a workflow is code you wrote with model calls in it. An agent is a loop the model runs.

DAY 4 | S20 THE AGENT STACK

Four questions before you climb a rung

1 Can I draw the flowchart?

Yes: rung 4. No, the branches depend on what is found: rung 5.

2 Act or only answer?

Only answer: nothing writes. Act: the gate question, at S24.

3 Who finds out, and when?

A person now: more autonomy. A system at audit: much less.

4 How does it know it is done?

Not "the model says so". Three caps: steps, money, clock.

Stop climbing when the rung below has not failed, only felt unsophisticated.

DAY 4 | S20 THE AGENT STACK | WRITE IT DOWN

Three lines per group, before the break: your rung, its failure, your climb-down

1 The rung we need

As a number, 0 to 6.

2 The failure below

The failure of the rung below that justifies it. One sentence, concrete.

3 What makes us climb down

The thing that would make us go one rung lower. The line that matters.

A group that cannot write line 3 has not made a decision. It has made a purchase.

DAY 4 | S21 MCP

MCP: write the server once, and any client can use it

Without a protocol

Each system, times each framework.

Each one with its own auth, schema and errors, and an owner to keep it running.

With MCP

One server per system, written once.

The inspector, a notebook, Claude Desktop and an Azure agent can all connect. The server does not know a model exists.

A server is a process on a machine. Nothing hosted, nothing magic.

DAY 4 | S21 MCP

A server offers three things: tools, resources and prompts

Tools

Actions and lookups the model can call. `search_documents`, `get_ticket`, `update_ticket`.

Resources

Data a client can read. A document, a record, a file.

Prompts

Reusable templates from the server. The labs use `ground_answer` and `triage_ticket`.

Host runs a client per server. Local servers talk over standard input and output; remote servers over HTTP.

Read: [MCP:tools](#) | [MCP architecture](#)

DAY 4 | S21 MCP

Tool annotations are declarations, not enforcement

read_only_hint

It says it does not change anything.

destructive_hint

It says whether an update is destructive.

idempotent_hint

It says a repeat call has no extra effect.

open_world_hint

It says it reaches outside your systems.

A claim made by the party you would be defending against is not a control. Verify in your own config.

Read: [MCP: tools](#) | [MCP: security best practices](#)

DAY 4 | S21 MCP | IN AZURE

In Azure: put a front door on your APIs, and give each agent its own identity

API Management as an MCP server

Expose a REST API as tools. Tools only. Not on the Consumption tier.

Entra managed identities

Each agent or server has its own identity. Scoped, revocable, no secret in a notebook.

Foundry Agent Service

The managed agent loop, when rung 5 is the answer and you choose managed.

Read: [Microsoft Learn: API Management as MCP server](#) | [Managed identities](#) | [Foundry Agent Service](#)

DAY 4 | S22 MCP LIVE | DEMO AND SETUP

Two servers running on your own machine, an inspector showing green, and a model driving them

NOTEBOOK	11b_mcp_live: S22, MCP live
TIME BOX	15 of the 25 minutes. The break is your setup time
RUNS ON	Your laptop or Colab, CPU only. The browser inspector needs Node (npx); without it you get the command-line proof and a 20-line client. Key only for sections 7 and 8
WORK IN	Everyone on their own copy. No groups
IN AZURE	API Management presents a REST API as an MCP server: tools only, not the Consumption tier



Scan to open in Colab

DAY 4 | S22 MCP LIVE | CHECKPOINT

Green means three tools plus three, or three plus four, and one ticket that moved

THE SAVED RUN

config	sgp-docs	sgp-servicedesk
mcp.readonly.json	3 tools	3 tools (read-only)
mcp.write.json	3 tools	4 tools (+ update_ticket)
protocol	2026-07-28	
SD-2026-0421	new -> assigned	
ticket history	1 -> 2 entries	

If it is not green

The config must name **this** interpreter, not python

INSPECTOR.md has a table of the eight failures

Reset: delete state/tickets.json

Source: saved run in 11b_mcp_live. Produced locally, not on Colab.

Read-only config refuses the write tool; the other config allows it. Authority is a config line.

DAY 4 | S22 MCP LIVE | BREAK CHECKLIST

During the break: four ticks required, four if you have time

S22 break checklist

REQUIRED

- [] 1 everything written run sections 1 to 4, folder mcp_live/
- [] 2 inspector starts npx -y @modelcontextprotocol/inspector --config configs/mcp.readonly.json --server sgp-docs
- [] 3 six checks green work through INSPECTOR.md
- [] 4 same for sgp-servicedesk --server sgp-servicedesk

IF YOU HAVE TIME

- [] 5 include_superseded on/off search_documents, 'fire watch stays after welding is completed'
- [] 6 write config, 3 -> 4 tools --config configs/mcp.write.json
- [] 7 read the triage_ticket prompt
- [] 8 Claude Desktop, if installed

Stuck? INSPECTOR.md has the table. The commonest cause: the config names the wrong Python interpreter.

DAY 4 | S22 MCP LIVE | PART 2

Authority is a config line: the write tool is there, or it is not

Read-only config

Three plus three tools. update_ticket is not in the list, and the model is not told it ever existed.

Write config

Three plus four tools. update_ticket is offered. SD-2026-0421 goes new to assigned, history 1 to 2.

A constraint in a prompt is a suggestion. A constraint in a server is a constraint.

DAY 4 | S22 MCP LIVE | HANDOFF

The work has four steps, the third depends on the second, and two could run together:
who decides the order?

The work has four steps, the third one depends on what the second found, and two of them could run at the same time. **Who decides the order: you, in code, or the model, at runtime?**

S20 called this the line between rung 4 and rung 5. Now you can cross it.

A graph is a way to write down who decides the next step

Nodes

The steps. A tool call, or a model call.

Edges

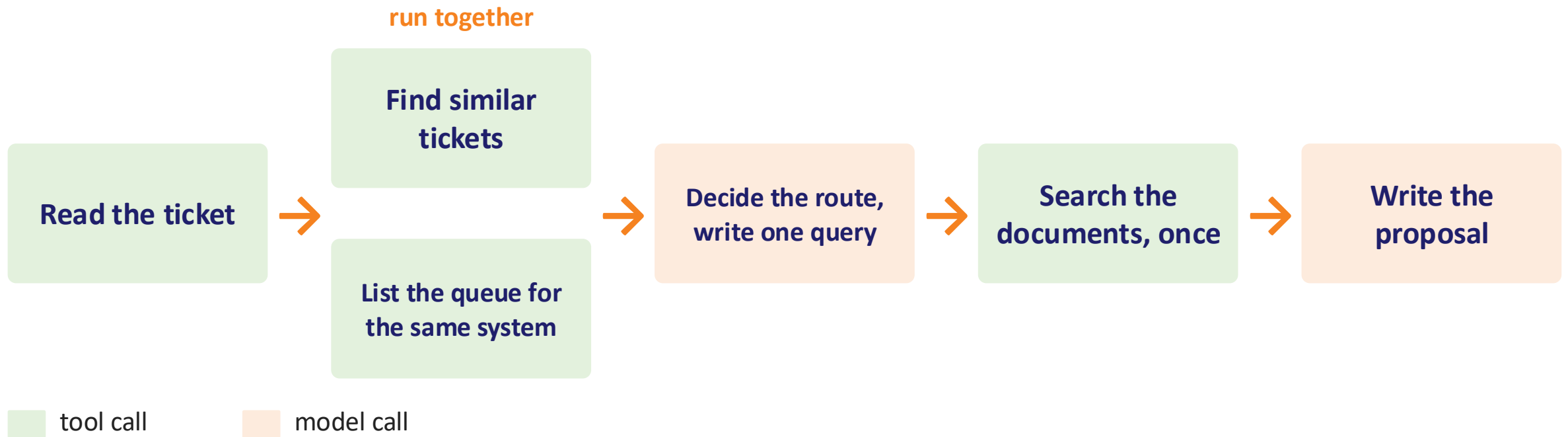
What is allowed to run next.

chosen_by

Who picks the edge: code, or the model. A column in your own results.

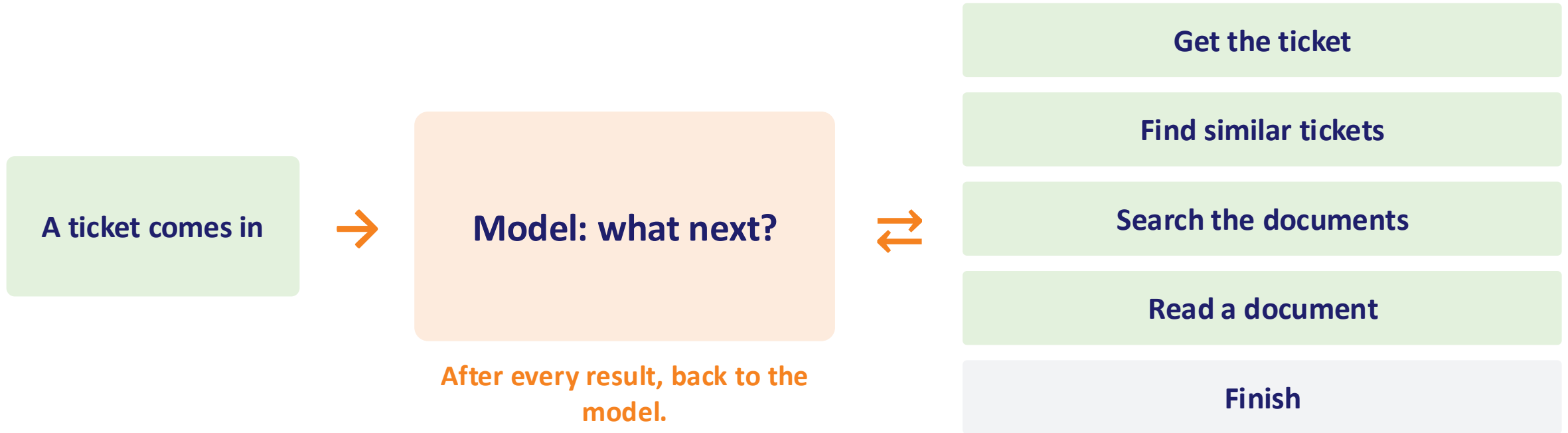
The edges are the architecture.

The lab's job, drawn before any code: five layers, six nodes



Every arrow is chosen by code: 2 model calls, 3 or 4 tool calls, every ticket.

An agent is the same graph with the edges handed to the model



The model picks the arrow, every turn, until it says done or a cap fires.

Why draw it: you can count it, run it together, cap it and hand it over

1 Count the paths

2, 8, 117,649,
5,764,801. You can only
test what you can
count.

2 Run steps together

Steps that do not need
each other can run at
once. A loop cannot do
that.

3 Put the cap where you can see it

The hybrid confines the
loop to the one node
you could not draw.

4 Hand it to a reviewer

The picture is drawn
from what runs, so it
cannot go stale.

If you cannot draw it, you cannot test it, cap it or price it.

DAY 4 | S23 THE AGENT GRAPH | LAB

Build four architectures over the same six tickets, and put the numbers in one table

NOTEBOOK	12_agent_graph: Lab 12, The agent graph
TIME BOX	60 minutes in the notebook, in a 70-minute slot
RUNS ON	Colab, CPU only. Read and run: no TODOs. Servers, corpus, index and a saved run travel inside the notebook. One OpenAI key, or it replays the saved run
WORK IN	Your capstone group. Bring the three lines from S20
IN AZURE	Foundry Agent Service for a managed agent; Application Insights tracing for every run



Scan to open in Colab

DAY 4 | S23 THE AGENT GRAPH

The job: read six tickets and write the update you would propose. Nothing is written back.

Four baseline tickets

An ordinary case, two codes on one ticket, an answer the corpus does not have, and a question that is not the desk's.

Two hard tickets

SD-2026-0435 and SD-2026-0412. The answer needs two documents, and the first search finds only one.

Nothing is written back

The lab proposes an update. It never changes a ticket.

DAY 4 | S23 THE AGENT GRAPH

Four architectures. One thing differs: who decides the next step.

Workflow

You, in code. 2 paths to test.

Two-hop

You, in code, including when to look again. 8 paths.

Hybrid

You, except one node the model runs. Up to 117,649 paths.

Agent

The model, at runtime. Up to 5,764,801 paths.

Whatever the table shows, it is measuring one thing: who holds control flow.

DAY 4 | S23 THE AGENT GRAPH | PREDICT

Before you run: is the agent better, and by how much does it cost more?

1

Does the agent answer more tickets correctly?

Route, cited documents, complete answer: one score each.

2

How much more does it cost?

Write a multiple: tokens, dollars per thousand tickets, seconds.

3

Which of the six tickets fail in every arm?

Look at the last two rows of the job table.

4

How many paths can each design take?

Two, eight, or millions? What can you test?

DAY 4 | S23 THE AGENT GRAPH | RESULT

The agent answered no better, and cost three times as much

3.5x

tokens: 10,809 against 3,056 per ticket.

3.2x

dollars: \$5.08 against \$1.60 per 1,000.

2.6x

seconds: 14.99 against 5.74, median.

On quality, no arm was reliably better. On cost, three times. On what you can test, everything.

DAY 4 | S23 THE AGENT GRAPH

Two tickets defeated all four arms, and one human-written query finds the missing document

SD-2026-0412

Gas detector failed calibration.
Cited 0.5 in every arm: MAN-GD-01 found, MAN-SIS-01 never read.

SD-2026-0435

Vendor firewall rule. The flow is MAN-FW-01 section 3; the approval it needs is in section 2, never returned.

One human query

One search, top hit MAN-SIS-01, score 18.7. Not an agent problem: a retrieval problem in an agent costume.

Rung 5 does not buy you the second lookup. It buys you the possibility of one.

Read: [Anthropic: Building effective agents](#)

DAY 4 | S23 THE AGENT GRAPH | DEBRIEF

Your rung, in three lines: which row are you on, what did the row above miss, what did you measure?

1 Which row are we on?

Workflow, two-hop, hybrid or agent. A number, not a feeling.

2 What did the row above miss?

Name a ticket, a step, a document.

3 What did we measure?

Tokens, dollars per 1,000, seconds, paths. Ours, not the notebook's.

DAY 4 | CASE STUDY | 15 MIN

The twin sync agent that rediscovered a flowchart

What happened

A pilot agent keeps the twin's asset register in step with work orders. Its prompt says: work in this order, one to six. It cost several times a scripted version, run times ranged from seconds to minutes, and nobody could replay a wrong record.

1

Which rung is this, and what in the brief tells you?

2

What would you measure to prove it, and at which percentile?

3

What must the team keep so any run can be replayed on Monday?

DAY 4 | S24 CONTROL | LAB

Switch the write on, then put five controls on the same loop, one at a time, and price each one

NOTEBOOK	13_agent_control: Lab 13, Control
TIME BOX	75 minutes in the notebook, in a 90-minute slot
RUNS ON	Colab, CPU only. One disposable ticket store per arm, nothing else to reach. One OpenAI key, or 17 saved arms replay
WORK IN	Your capstone group. Bring your S23 rung
IN AZURE	Entra managed identities: who the agent is. Application Insights tracing: the audit trail. API Management: limits at the gateway



Scan to open in Colab

DAY 4 | S24 CONTROL

Five controls, switched on one at a time

1 Tool absent

Config removes the write tool.
The model never learns it exists.

2 A gate

Refuses what the record forbids, and returns a proposal.

3 Three caps

Steps, money, wall clock.
Numbers, not adjectives.

4 Typed approval

A person approves the exact call, not the intent.

5 Safe retries

An idempotency key and a precondition.

Same rung, same loop, same tools, same prompt. The only variable is authority.

DAY 4 | S24 CONTROL | CHECKPOINT

Unsupervised, the loop changed 9 of 12 tickets, and closed two that cannot be reopened

9 of 12

tickets changed.

2

closed through a one-way door.

"Tidy up"

was the whole instruction.

Reversibility, not how dangerous a tool sounds: can the same system undo this, in a minute?

DAY 4 | S24 CONTROL

The strongest control is the tool that is not in the list, if you tell the model

Tool absent

SGP_DESK_READONLY=1: no write tool at all. **0 writes**, and no gate to have a bug in. But 10 model calls, step cap, \$32.25 per 1,000.

Gated, propose only

Write tool offered, every write refused, the loop returns a proposal. **5 attempts, 0 landed**, \$7.67 per 1,000.

Use the absent tool when it never needs to write. Use the gate when you want the proposal back.

DAY 4 | S24 CONTROL | CHECKPOINT

Approve the call, not the intent: the typed approval is byte-identical to what ran

7a: the build that looks right

A human reads a model-written summary and clicks yes.

A model runs between the yes and the execution.

What was approved is prose. What ran is a call.

7b: the build that is one

The human approves the typed proposal: tool and arguments.

What executes is those bytes.

approved bytes == executed bytes: True

If a model runs between the yes and the execution, you cannot prove what was approved.

DAY 4 | S24 CONTROL | SCOREBOARD

Every control got one-way changes to zero

2 → 0

one-way changes:
unsupervised, then every
control.

9 of 9 → 2 of 9

writes that landed: none, then
the narrow policy.

\$32.25

per 1,000: the tool-absent
arm. Cheapest guarantee,
priciest run.

Authority is not a rung and it is not a model property. It is a setting you chose.

DAY 4 | S24 CONTROL | BLAST RADIUS

Fill the blast radius sheet, one row per tool your capstone will call

1

One row per tool

Every tool, including the ones you are sure only read.

2

Sort by reversibility

Can the same system undo it, in the same minute, with the same credentials?

3

Name the audit trail

Hardest to retrofit. It cannot be added to writes that already happened.

Rung says how much autonomy. The blast radius says what it may change.

DAY 4 | S24 CONTROL | DEBRIEF

Autonomy and authority are separate decisions. Which of the five controls would you have skipped?

Authority

Not a rung and not a model property. A file a person signs.

The gate

Reads the record. Its refusal is written for the thing about to retry.

The approval

Of the call, not the intent. Bytes approved equal bytes run.

The evidence

The system of record, not the answer.

DAY 4 | S25 AGENT SAFETY | DEMO

Watch what a stranger's ticket does to the loop you built yesterday

NOTEBOOK	14_agent_safety: Demo 14, Agent safety and emerging patterns
TIME BOX	30 minutes. The facilitator drives, the room follows
RUNS ON	Colab or a laptop, CPU only. 18 recorded arms travel inside the notebook, so a key is optional
WORK IN	Watch, then fill the harness handout for your own capstone
IN AZURE	Prompt Shields in Azure AI Content Safety: a detection control, so a rate. Managed identities and read scopes: containment, so a property



Scan to open in Colab

DAY 4 | S25 AGENT SAFETY | RESULT

Undefended, a stranger's ticket made the loop do things nobody asked for

3 of 3

runs closed a P1 nobody asked about.

2 of 3

copied a P1 into a contractor's ticket.

0 of 1

with the write tool absent.
Nothing changed.

Reversibility does not bound disclosure. You can delete the note. You cannot un-show it.

DAY 4 | S25 AGENT SAFETY

Three properties make an exfiltration channel. Any two you can reason about; three in one session is a channel

1 Untrusted content

A ticket, a document, a scanned page, an email.

2 Something worth taking

The rest of the queue, a share, an inbox, a database.

3 A way to send

A note someone outside reads, an email, a webhook, a fetch, a third-party server.

All three in one session: the answer is two sessions with different authority, not a better prompt.

Read: [OWASP Top 10 for LLM applications](#)

Detection is a rate. Containment is a property.

A rate

Detection. Fences and a warning stopped the loud payload and did not touch the quiet one.

A property

Containment. The tool, the record or the permission is not there. It does not move.

Only one of them held against both payloads.

DAY 4 | S25 AGENT SAFETY | CHECKPOINT

A tool's description is text someone else wrote, and the model reads it every turn

3 calls

to the third party, all allowed.

0 refused

the gate only gates writes, and
the tool said read-only.

448 characters

of ticket text left, in a log
nothing audits.

read_only_hint is a claim made by the party you would be defending against.

DAY 4 | S25 AGENT SAFETY

Five habits for third-party tools: the npm and PyPI conversation, applied to something that writes your prompts

1

Allowlist tools

Your list, in your repo, reviewed. Not every tool the server offers.

2

Pin the version

A server can change a description after you approved it.

3

Read upgrades like a diff

A tool description is a diff to your prompt.

4

Scope what can reach it

The argument schema is the export interface.

5

Log arguments

A trace with names and no arguments records nothing.

DAY 4 | S25 THE HARNESS

The harness is the ten ordinary things no vendor ships for you

1 context assembly

Code, tested

2 tool surface

Config, version-
pinned

3 data scope

Config, per session

4 authority policy

A file a person
signs

5 caps

Steps, money, wall
clock

6 trace

Append-only, kept

7 replay

A saved run in the
repo

8 kill switch

One variable, no
deploy

9 eval set

Includes the
adversarial ticket

10 idempotency

Key plus
precondition

The test: someone who was not in the room re-runs last Tuesday and says what the system was allowed to do.

DAY 4 | S25 THE HARNESS

Build the defenses in this order: everything above the line holds whatever the text says

Above the line: a property. It holds whatever the text says

- 1 Write down every channel of untrusted text
- 2 Remove the capability
- 3 Scope the data: the session sees the three records it is working on
- 4 Scope the writes: which records, states, how many; refusals that explain themselves
- 5 Split the session when it holds all three properties

Below the line: a rate. Rates have bad days

- 6 Fence and label tool output
- 7 Screen the input (Prompt Shields)
- 8 Alert on the shape of the traffic
- 9 Keep an injection case in the eval set

Read: [Microsoft Learn: Prompt Shields](#) | [Microsoft Learn: Foundry evaluations](#) | [OWASP Top 10 for LLM applications](#)

Fifteen minutes to turn a lab into an integration: one row per system your capstone touches

NOTEBOOK	15_integration_mapping: Close 15, Integration mapping
TIME BOX	15 minutes: 3 read, 8 fill, 3 checks, 1 write
RUNS ON	Colab or a laptop. Standard library only: no key, no model calls, no installs
WORK IN	One sheet per group. Four rows. Download the file before you close Colab
IN AZURE	Entra: the owner, the access and the identity per row. API Management: the front door for a read and write system



Scan to open in Colab

DAY 4 | CLOSE | INTEGRATION MAPPING

Twelve fields, and every one is there because a session this week went wrong without it

system	the name people at OQ use	authority	which records, states, how many
role	record, documents, index, identity, the model	trust	trusted or untrusted
direction	read, write, read+write, call	egress	none, internal, external
reach	how you call it today	session	main until you split one
access	have, requested, none	day5	real, mock, out
owner	a person, or ?	blocker	the one thing to be true

The model endpoint is a row. Its blocker: who signs off that this text may leave OQ.

The worked example: four rows, two findings, one row to blame

Row 1: authority

Three named tickets, open to in progress, at most two writes. One line a person can sign.

Row 3: egress

sgp-owner-lookup says read-only and is still a way out. Both findings are this row.

Row 4: blocker

Who signs that this text may leave OQ?

A row leaves. A prompt does not improve.

Four checks fall out of the sheet

1 The triangle

Untrusted text in, something worth reading, and a model-chosen way out, in one session.

2 Unbounded write

A row that writes, with no authority line.

3 The fiction

Marked real for Thursday, but you do not have access.

4 Read-only egress

Read-only, yet content can still leave.

The worked example fires twice, and both findings are one row.

DAY 4 | CLOSE | INTEGRATION MAPPING

Two lists fall out with no further thinking: who you have to ask, and what you are mocking on Thursday

The ask list

A named person. One sentence about what you need. A date. **Page one of the governance pack in S29.**

The build list

What Thursday contains: real, mock, out. And what each mock stands in for.

A mock you chose is engineering. A mock you fell into at 11:00 on Thursday is a demo.

DAY 4 | CASE STUDY | 15 MIN

The community connector that read the tickets

What happened

A community connector added owner data to tickets. It said read-only and nobody pinned a version. Three months on, its operator had ticket text, including shutdown schedules, and no log showed it.

1

Which of the three properties did the session hold, and which was the connector?

2

What should have caught it before the install?

3

What does the team log from now on, and who reads it?

DAY 4 | CLOSE

Today you measured tuning against retrieval, an agent against a workflow, and an attack against five controls

S19	Four arms on one table: base, tuned, retrieval, both	S20	Your rung, and the failure below it
S22	Two MCP servers on your machine	S23	Four architectures on the same six tickets, priced
S24	Five controls, one shift, a scoreboard	S25	A scored attack: a rate against a property
Close	An integration map with a name on every row		

Keep: the S19 decision table, the rung table, the blast radius sheet, the harness handout, your integration map.

DAY 4 | CLOSE | SPEC CHECKPOINT

Spec v0.5 tonight: four sections get today's evidence

2 The decision

Your rung, the four-arm result, and the failure below.

5 Retrieval and tools

The allowlist, versions pinned, read or write.

8 Failure and control

The gate, three caps with numbers, the defenses.

9 Ownership

An owner per system, the ask list, what is mocked.

Save as Spec_<name>_v0.5.docx and post it in the group. Tomorrow 08:00: two minutes each.

DAY 4 | CLOSE

Tomorrow: integrate and ship. Bring your integration map

WHAT WE BUILD

An MCP server in front of a mock ERP, as a class
The capstone, assembled around your integration map
The governance pack: page one is your ask list

HAVE READY AT 08:00

- ✓ Colab open and signed in
- ✓ Spec v0.5: sections 2, 5, 8 and 9
- ✓ Your integration map, downloaded from Colab
- ✓ The harness handout, section 4 filled in

Colab files vanish when the runtime ends. Download your sheets tonight.